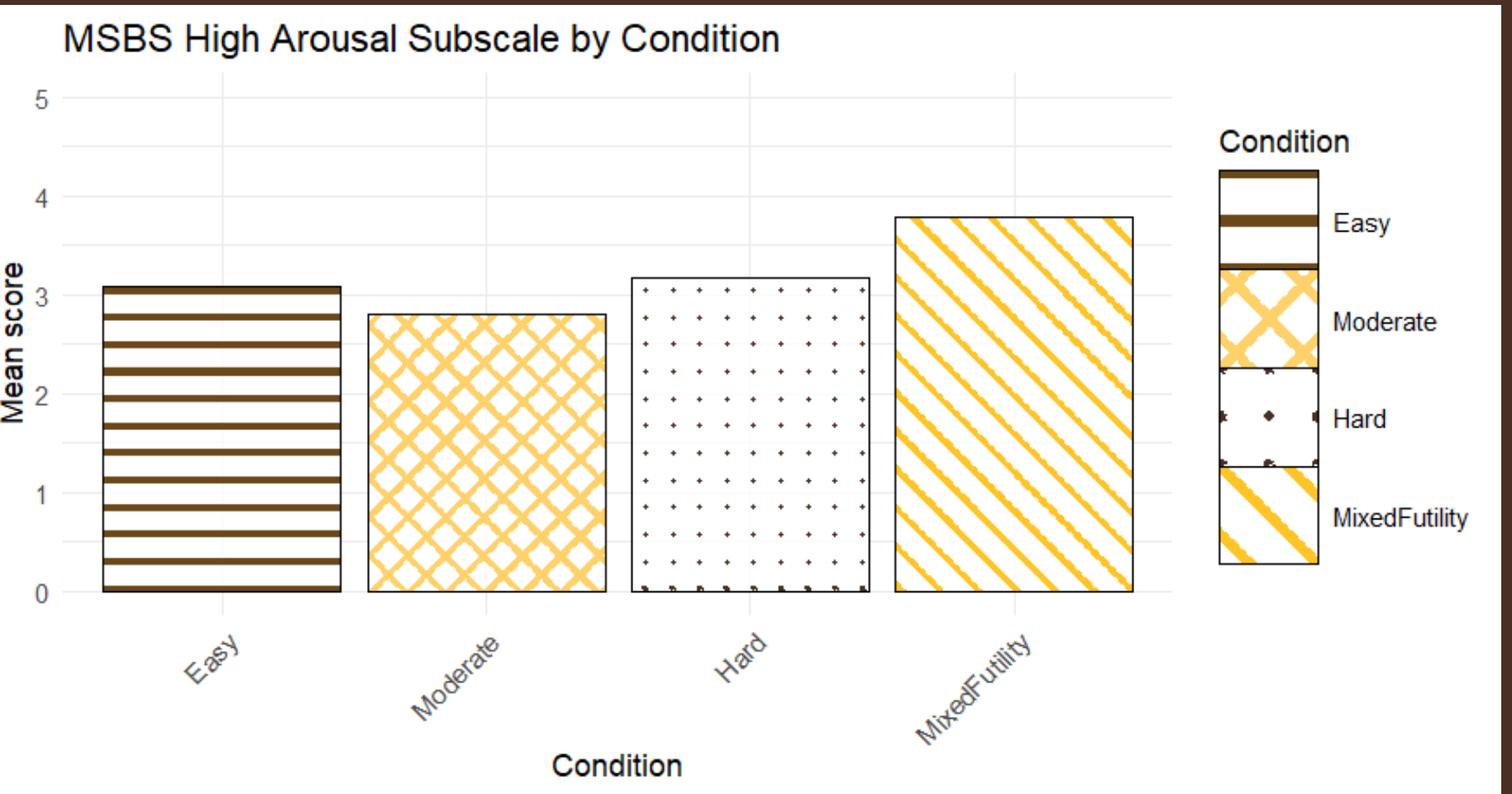
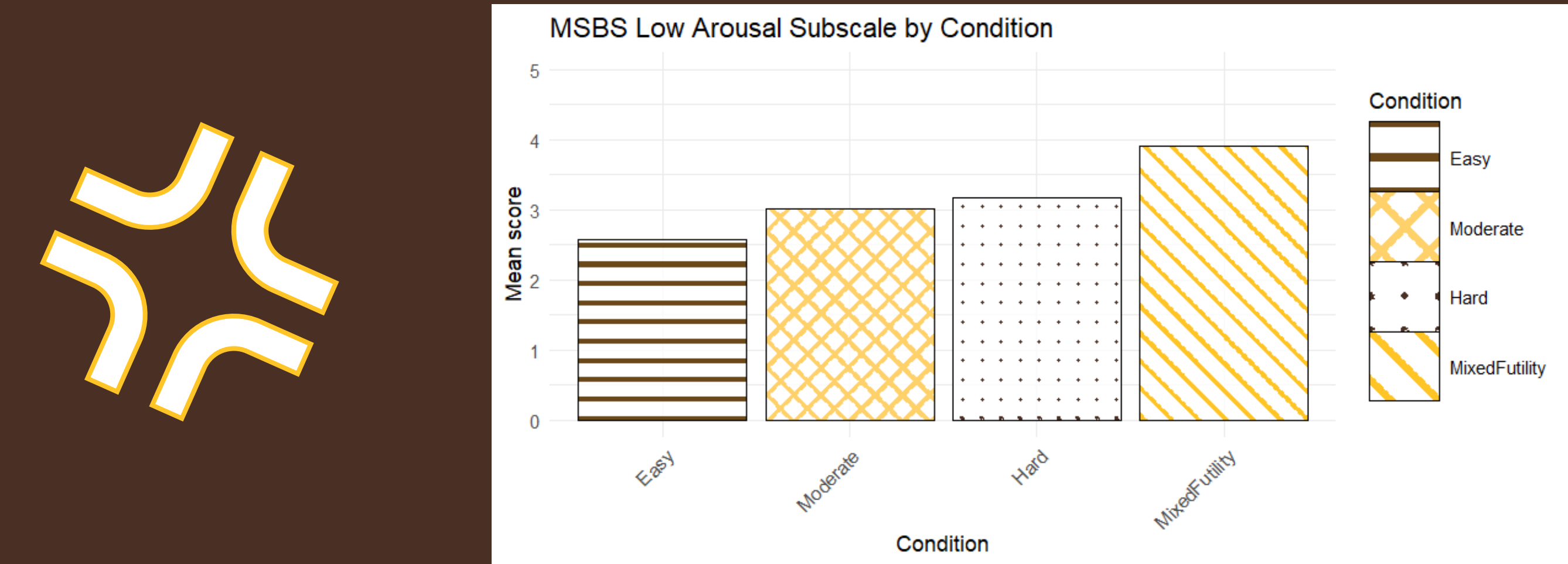
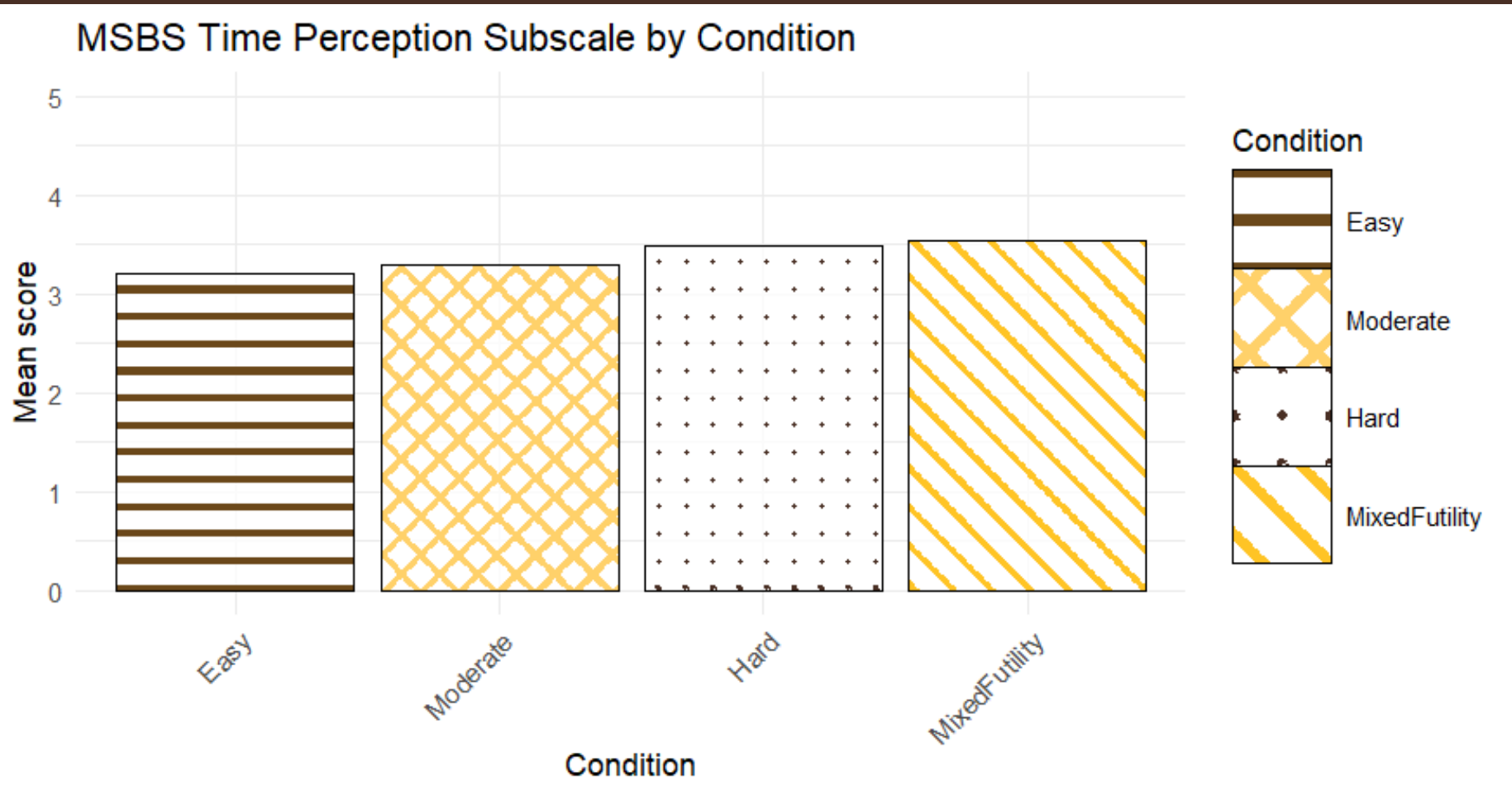
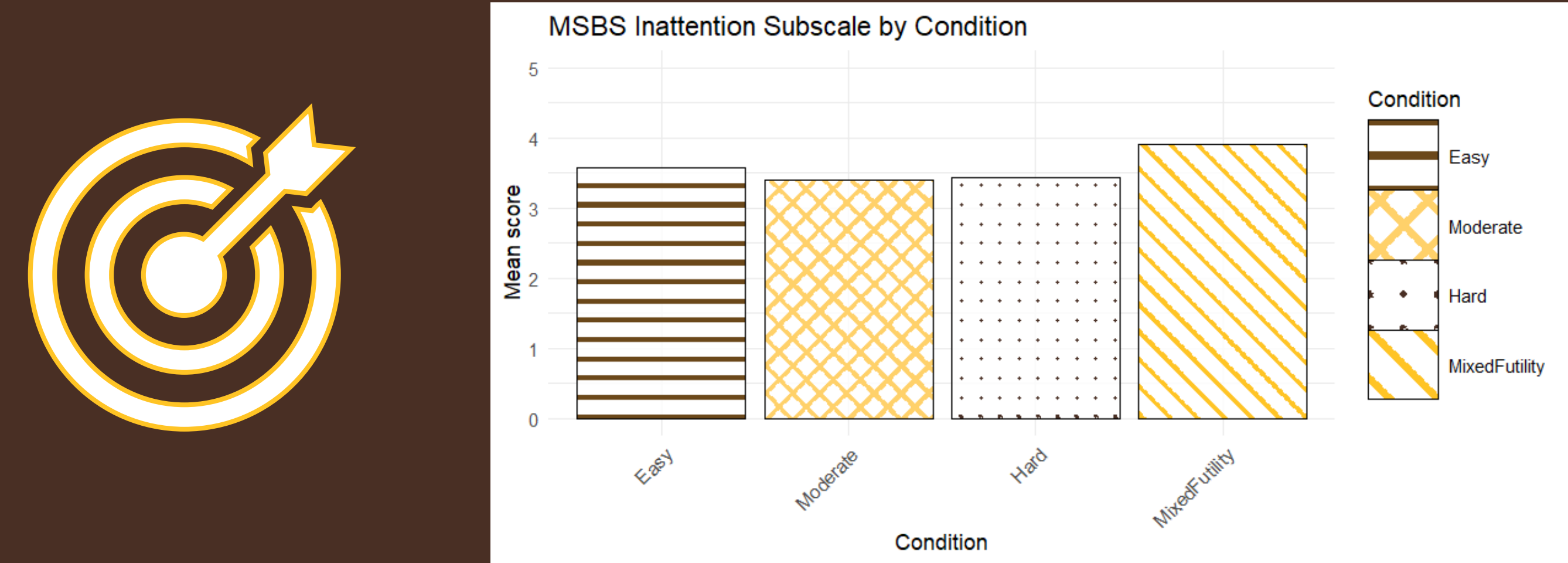
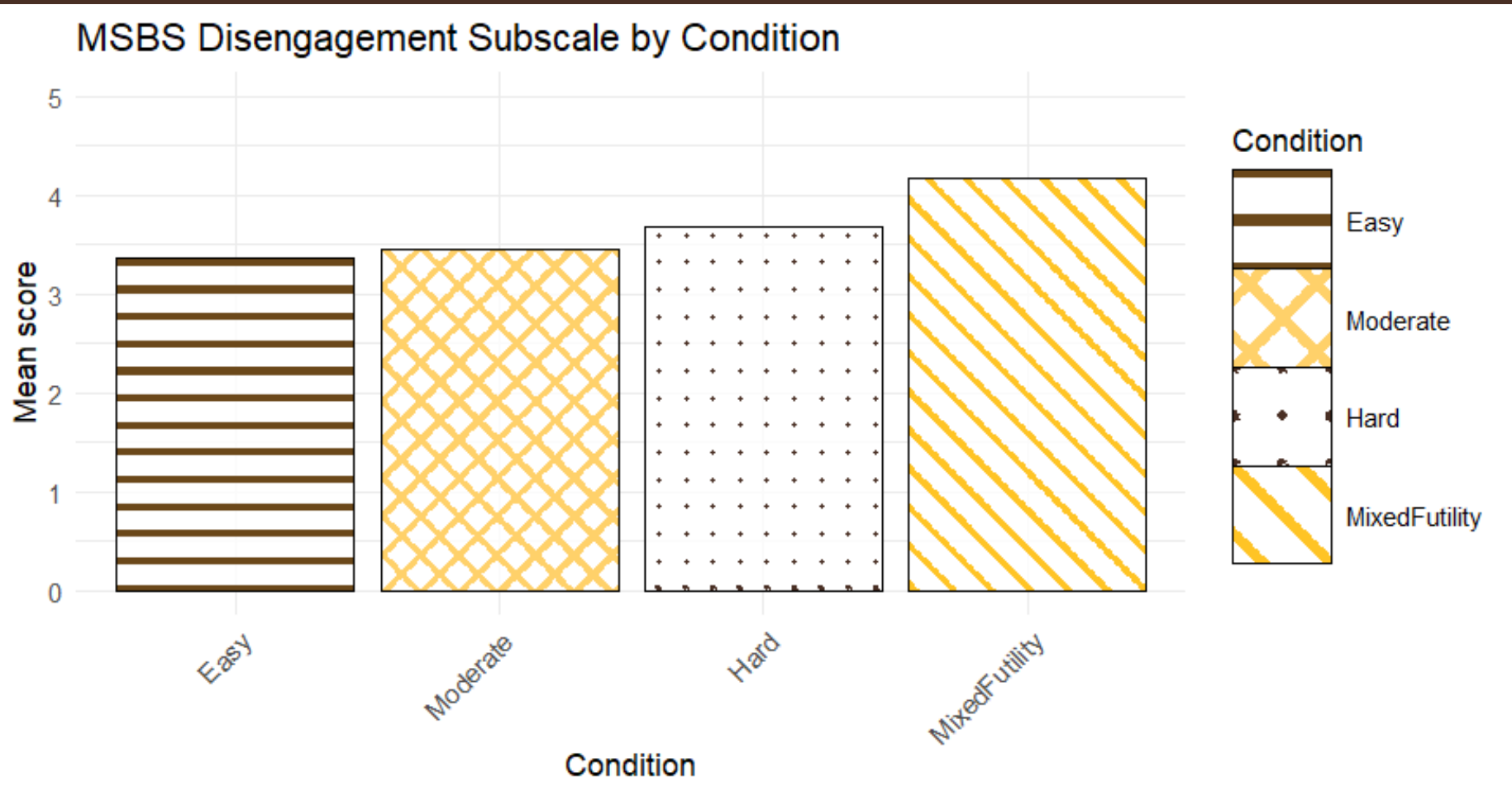


Targeted Boredom: Developing an Anagram Battery to Induce Distinct Profiles of Boredom

Matthew R. Helm



Method

Participants

- Adults (18+) recruited via Cloud Research’s Connect Platform and the University of Wyoming’s research participant pool (SONA Systems).
- Current Sample: $N = 149$ (target ≈ 200 ; data collection ongoing)
- Age: $M = 38.09$, $SD = 15.36$.
- Race/Ethnicity: White $\sim 73.25\%$, Black $\sim 9.55\%$, Asian $\sim 9.56\%$, Latine $\sim 4.46\%$, Indigenous, Middle Eastern, Multiracial, and Undisclosed $\sim 3.18\%$.
- Sex/Gender: Women $\sim 65.77\%$, Men $\sim 32.89\%$, Nonbinary $\sim 1.34\%$

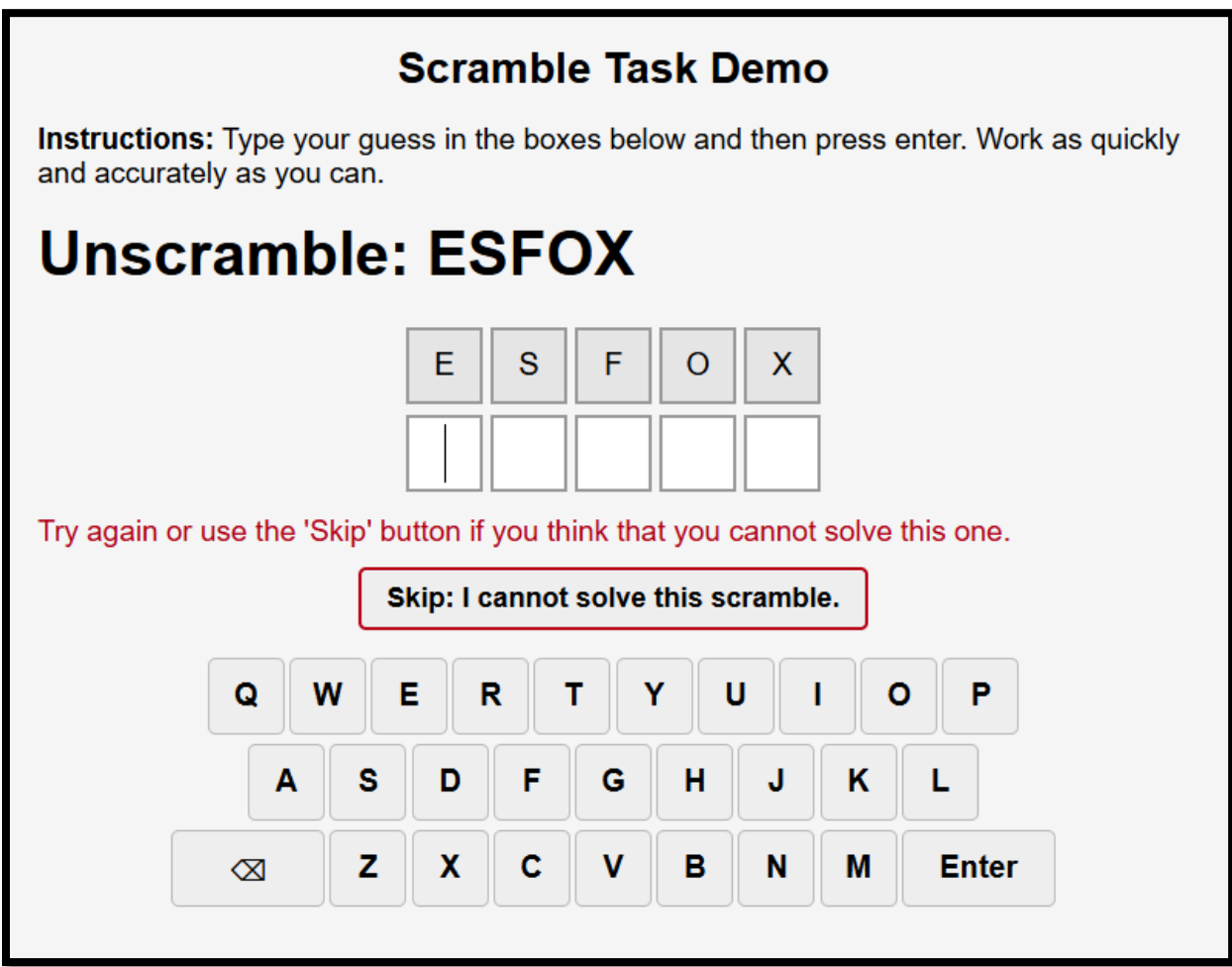
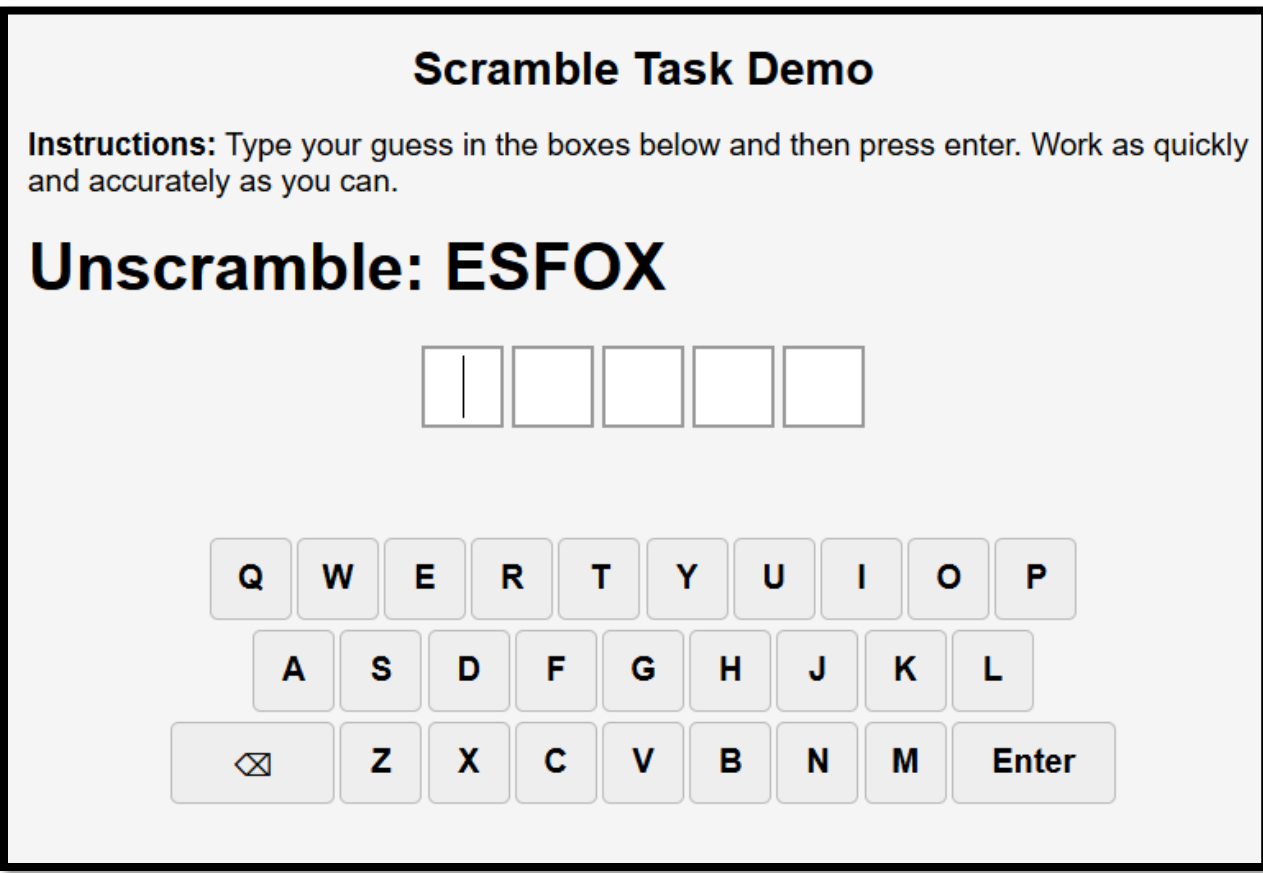
Materials

- Anagram set: 200 five-letter English words with unique letters, derived from Gilhooly and Hay’s norms and cross-referenced with the SUBTLEX corpus.
- For each solution, all 5-letter permutations ($5! = 120$ scrambles) were generated and then scored on four indicators of scramble difficulty.
 - First Letter Match* (scramble’s first letter matches first letter of solution).
 - Last Letter Match* (same logic as First Letter Match, but for the last letter).
 - Bigram Frequency* (weighted by natural-language bigram frequencies; higher = more word-like and predicted to be easier to solve).
 - Levenshtein Edit Distance* (more numbers of edits are harder to solve).
- A composite difficulty index (z-scored and weighted) was calculated for each scramble from the scramble difficulty indicators using the following formula: $(2 * FLM) + (2 * LLM) + (0.5 * BigFreq) + (-1 * LED)$
- Scrambles were selected according to the following:
 - Easy = Highest composite score (easiest to solve)
 - Moderate = Average composite score.
 - Hard = Lowest composite score (hardest to solve).

- Per word, three scrambles were selected to represent Easy, Moderate, and Hard difficulty levels, a separate list was created to include 25% unsolvable items in Mixed Futility to account for possible learned helplessness effects.
- Task delivered via a custom web-based anagram interface (hosted on GitHub) with trial-level logging (solution, scramble, condition, RT, accuracy, etc.) parsed to Qualtrics (with the study measures).
- The Multidimensional State Boredom Scale (MSBS) which consists of five subscales rated on a seven-point agreement scale from 1 = Strongly Disagree to 7 = Strongly Agree:
 - Disengagement/Lack of Meaningful Engagement* (e.g., I feel like I am sitting around waiting for something to happen).
 - Inattention/Unfocused* (e.g., It is difficult to focus my attention).
 - High Arousal/Frustration* (e.g., I feel annoyed with the people around me).
 - Low Arousal/Apathy* (e.g., I feel down).
 - Time Perception/Slowing* (e.g., Time is moving very slowly).

Procedure

- Consent and task instructions preceded a 15-minute solving block for the assigned scramble difficulty condition.
- Participants were randomly assigned to Easy, Moderate, Hard, or Mixed Futility condition (conditions varied by scramble difficulty and presence/absence of unsolvable trials).
- During the solving block, participants saw one scrambled 5-letter string at a time and attempted to solve as many as possible. They could also skip trials if they believed the scramble they were presented with was unsolvable.
- After the solving block, participants completed the MSBS, manipulation (i.e., “how difficult was the task”) and attention checks, exploratory measures (e.g., Intrinsic Interest for tasks, perceived meaningfulness), and demographic questions.



Introduction

- Boredom is an aversive state that is linked to a wide range of negative outcomes (e.g., impulsivity, poor self-control, risky behavior).
- The Meaning and Attentional Components (MAC) model proposes that boredom arises from a mismatch between a task and one’s valued goals (i.e., meaningfulness) or from a mismatch between attentional demand for a given task and one’s cognitive resources.
- On the attentional dimension, boredom can reflect either:
 - Understimulation*: low demand, excess resources (low-arousal, “flat” boredom).
 - Adequate/Optimal Engagement*: interest, enjoyment (a lack of boredom).
 - Overstimulation*: high demand, insufficient resources (high-arousal, frustrated boredom).
- Prior work has used anagrams to induce boredom, but there is no standardized anagram battery specifically designed to produce distinct boredom profiles (e.g., understimulation vs engagement vs overstimulation).
- The current project aims to develop an anagram battery that manipulates scramble difficulty to elicit the distinct boredom profiles from the MAC model of boredom in order to help individuals possibly afflicted by the negative downstream consequences of different types of boredom.

Results

Table 1

Average Task Performance Across Scramble Difficulty Conditions

Condition	RT (sec)	Trials (n)	Attempts (n)	Skips (n)	Accuracy (%)	Difficulty
Easy	14.23	92.7	1.19	8.84	90.6	3.26
Moderate	23.58	46.1 ^b	1.55 ^c	9.69 ^d	82.7 ^a	4.24 ^e
Hard	23.73	54.4 ^b	1.57 ^c	12.2 ^d	82.2 ^a	4.35 ^{ef}
Mixed Futility	29.97	39.9 ^b	1.88	16.9	63.7	4.84 ^f

Note. Superscripts that share the same letter are not statistically significantly different from one another.

Table 2

Average MSBS Subscale Scores Across Scramble Difficulty Conditions

Condition	Disengagement	Inattention	Time Perception	Low Arousal	High Arousal
Easy	3.36 ^a	3.57 ^b	3.2 ^c	2.58 ^e	3.09 ^g
Moderate	3.45 ^a	3.41 ^b	3.3 ^c	3.01 ^{de}	2.8 ^g
Hard	3.69 ^a	3.44 ^b	3.5 ^c	3.17 ^{de}	3.18 ^g
Mixed Futility	4.17 ^a	3.91 ^b	3.54 ^c	3.91 ^e	3.78 ^g

Note. Superscripts that share the same letter are not statistically significantly different from one another.

Discussion

- Data collection is still ongoing so results should all be taken with that in consideration. Data also have not been fully screened for bad actors (e.g., those who failed attention checks, or may not have completed the task as instructed).
- Contrary to predictions first solution accuracy between Moderate and Hard did not differ. Easy’s percentage of solution accuracy was right at 90% (as predicted), but all other conditions were higher than predicted.
- While Mixed Futility did seem to produce the overstimulated boredom profile (as measured by high scores on the MSBS High Arousal and other boredom predictors), Hard and Moderate rarely differed (as was predicted), suggesting that hard wasn’t quite difficult enough to induce the overstimulated profile of boredom (which was also predicted).
- Easy, while definitively the easiest scramble condition (per the difficulty manipulation check), did not seem to produce the understimulation form of boredom that was hypothesized. It’s plausible that the minor inconvenience of a simple scramble was still too interesting/entertaining to induce the predicted type of boredom. Future directions might consider making the easy scrambles even easier (possibly inducing a Mixed Pre-Solved condition).



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